A STUDY ON URBAN MOBILITY

Delhi Traffic Analysis

Patterns, Congestion Hotspots & Recommendations for a Smarter Commute

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INTRODUCTION

Introduction to the Study

Delhi, one of the world's largest and most densely populated metropolitan regions, faces mounting pressure on its road network every day.

Rapid urbanization, rising vehicle ownership, and limited road expansion have made traffic congestion a persistent challenge for commuters, businesses, and city planners alike.

This project analyses traffic patterns across the city to identify congestion trends, peak-hour behaviour, and key problem areas — with the goal of suggesting practical, data-backed improvements.

33M+

Population in NCR

1.4Cr+

Registered Vehicles

Top 10

Most Congested Cities Globally

Figures are illustrative estimates used for classroom discussion.

PROBLEM STATEMENT

Why This Study Matters

1

Rising Vehicle Numbers

Vehicle registrations continue to outpace road capacity growth year after year.

2

Peak-Hour Gridlock

Morning and evening rush hours see travel speeds drop sharply on major corridors.

3

Economic & Health Cost

Congestion wastes commuter time, fuel, and worsens air quality across the city.

4

Inadequate Public Transit Reach

Last-mile connectivity gaps push more commuters toward private vehicles.

OBJECTIVES

Project Objectives

1. Study traffic volume and flow patterns across major Delhi corridors
2. Identify peak congestion hours and high-density traffic zones
3. Analyse causes behind recurring bottlenecks and delays
4. Evaluate the role of public transport in easing road traffic
5. Recommend actionable, data-driven solutions for smoother mobility

METHODOLOGY

Scope & Methodology

1
Define Scope

Focus on major arterial roads & intersections in Delhi NCR.

→

2
Collect Data

Gather traffic counts, speed data & commuter survey responses.

→

3
Clean & Organise

Structure raw data into time-based and location-based sets.

→

4
Analyse Patterns

Study peak hours, congestion hotspots & contributing factors.

→

5
Report Insights

Translate findings into practical recommendations.

DATA & TOOLS

Data Sources & Tools Used

Data Sources

- Traffic Police / Government open-data portals
- Google Maps & navigation-app traffic layers
- Field observation counts at select intersections
- Commuter survey responses (sample group)

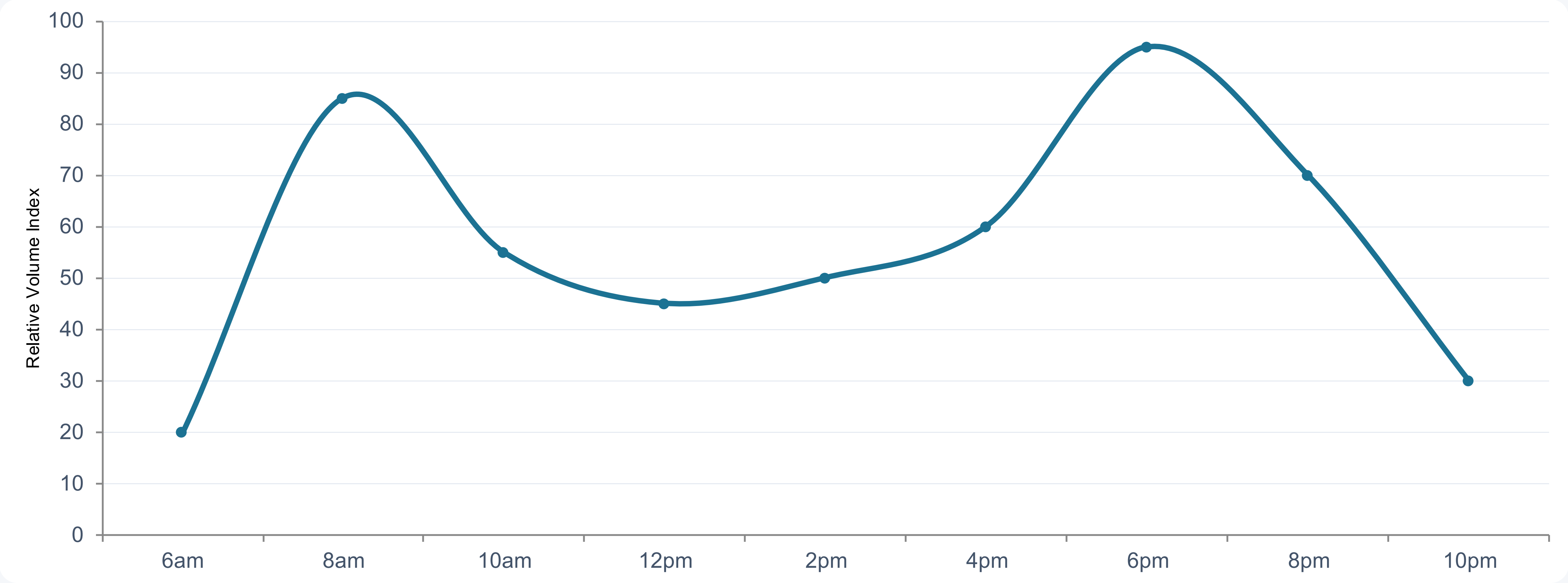
Tools Used

- Microsoft Excel — data cleaning & pivot analysis
- Charts & dashboards — visual trend analysis
- Google Maps traffic view — hotspot mapping
- Survey forms — commuter perception data

DATA ANALYSIS

Traffic Volume Through the Day

Illustrative hourly traffic-volume trend showing clear morning and evening peaks.



KEY FINDINGS

Major Congestion Hotspots

ITO Junction

Severe delays during both peak windows

Ring Road (AIIMS–Dhaula Kuan)

Heavy commercial + commuter overlap

NH-48 towards Gurugram

Bottleneck at city border crossings

Mukarba Chowk

High inter-state freight traffic

Connaught Place ring

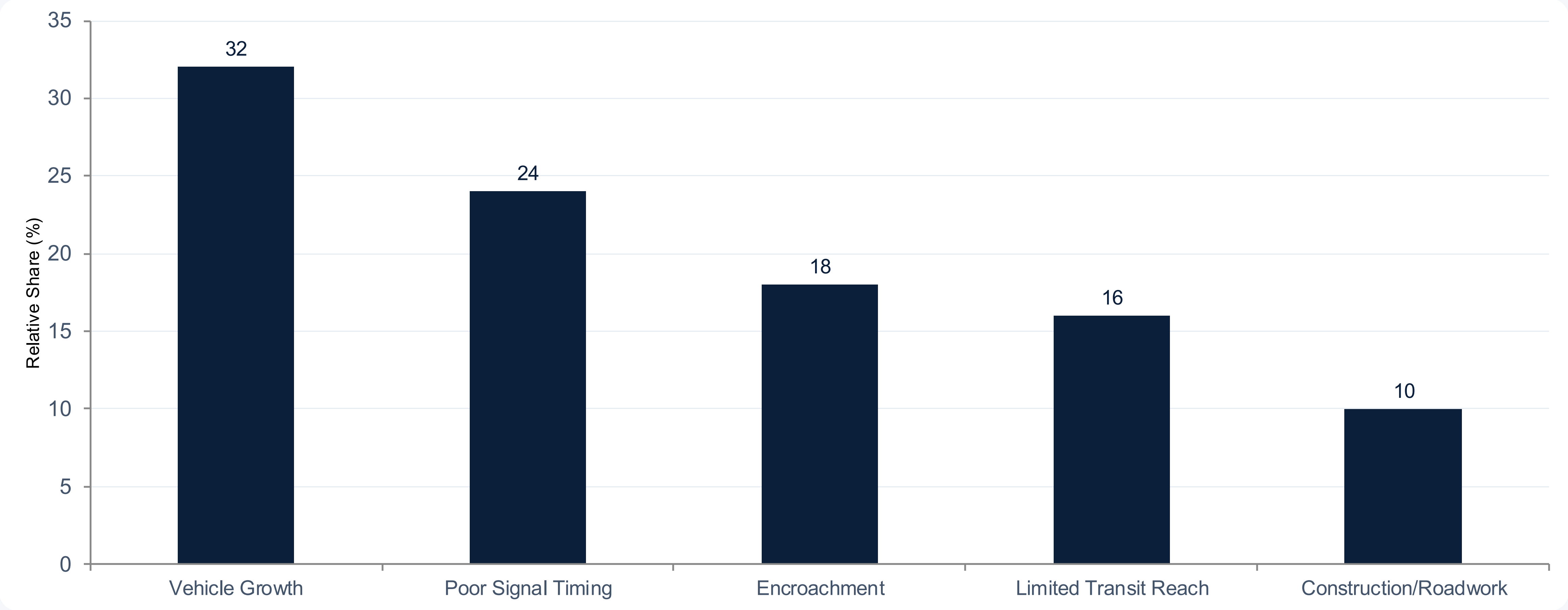
Dense mixed traffic, limited lanes

Akshardham–Noida Link Road

Peak-hour cross-river congestion

KEY FINDINGS

Factors Contributing to Congestion



RECOMMENDATIONS

Suggested Solutions

1

Smart Signal Systems

Adaptive traffic signals that respond to real-time flow at major junctions.

2

Strengthen Last-Mile Transit

Expand feeder buses & shared mobility around metro stations.

3

Staggered Office Hours

Spread out peak demand across offices, schools & institutions.

4

Dedicated Bus & Cycle Lanes

Reallocate road space to move more people, not just more vehicles.

5

Real-Time Traffic Apps

Encourage route planning around live congestion data.

6

Stricter Encroachment Control

Keep carriageways and footpaths clear for smoother flow.

Conclusion

- Delhi's traffic challenges stem from a combination of rapid vehicle growth, uneven infrastructure development, and gaps in public transit reach.
- Clear peak-hour patterns and recurring hotspots show that targeted interventions — smarter signals, better transit connectivity, and demand management — can meaningfully ease congestion.
- A combination of technology, policy, and citizen behaviour change offers the most sustainable path toward smoother, safer mobility in the city.

Thank You

Questions & Discussion Welcome

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